TFPT — Safeguards against Coincidence and Numerology

The verification discipline: how every load-bearing claim is defended against chance, fitting and over-reading

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions** (plus two standalone working notes), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** `tfpt_prime_front` — research documentation of the prime / zeta line: the seven load-bearing modules on the E_8 glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8,4,4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified); **N2.** `note_hilbert_polya_truncations` — a computable, zeta-free truncation family for the Weil measure: measurements on a Hilbert–Pólya candidate (prediction-freeze methodology; no RH claim).

You are reading the ***Safeguards*** *companion (S): it collects, in one place and in one language, every mechanism by which TFPT protects a claim against coincidence — the status calculus, the anti-fitting rules, the over-determination accounting, the firewall, the frozen predictions and null model, the two independent proof paths (Wolfram, Lean), the red team, and the honest residual. It introduces no new physics; it makes the discipline auditable.*

Abstract

A two-input theory that reads out many small integers is, a priori, at risk of being elaborate numerology. TFPT answers that risk not with rhetoric but with a *layered, machine-checked*

discipline, and this companion states all of it uniformly. The layers are: (1) a four-class status calculus with a single-source ledger and a sync audit that forbids a conditional claim from being rendered as exact; (2) an anti-fitting rule ("no free pattern") plus a reverse audit that counts how much structure carries *no* readout; (3) an over-determination map that honestly separates *multiplicative* evidence from *compression* (many readouts from one generator) and, applied to TFPT's own arithmetic witnesses, finds them to be facets of *one* $(2, 3, 5)/E_8$ object (compression), locating the genuine multiplication in an input forced four independent ways plus a foreign readout (α^{-1}); (4) a firewall that keeps external-physics interfaces from ever being typed as compiler outputs, together with the No-Unit theorem that makes the absence of an absolute scale a *theorem*; (5) a frozen prediction registry with a Monte-Carlo null model and a live data scorecard; (6) two independent reproduction paths — an independent Wolfram engine and a Lean 4 kernel proof; (7) an adversarial red-team layer; and a final, explicit statement of what would still falsify the theory. The thesis is deliberately modest: these safeguards make coincidence an *expensive* explanation of the discrete core, and they keep "exact compiler closure" from ever being mistaken for "closed physics".

1 The risk, and the shape of the answer

The honest danger for TFPT is not that an individual number is wrong — the arithmetic is usually clean — but *closure inflation*: that an exact, discrete *compiler* closure is read, or worded, as a *physical* closure. Every safeguard below exists to hold that one line. None of them, alone, turns syntax into semantics; together they make the discrete core hard to dismiss as chance and impossible to oversell as a finished Theory of Everything. We state each as a *layer* with its enforcing artefact(s).

Figure 1: The seven safeguard layers; a load-bearing claim must pass all of them. Each layer names its enforcing artefact (script id or document).

2 Layer 1 — the status calculus (no `[C]` dressed as `[E]`)

Every claim carries one of four display markers: [E] exact / proven (identity, Lie/lattice theorem, Lean-formalised, or numerical fixed point), [C] conditional (physical, under named hypotheses), [O] open / axiom / anchor, [X] falsifiable kill test. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) lives in the *single source of truth*, `verification/status_ledger.csv`; the papers and the website only *mirror* it. A sync audit (`verification/audit_sync.py`, run as `bash build.sh audit`) enforces, in both directions, that the suite, the runner, the registry and the ledger agree; that every script is cited in a paper *body*; and that no generated surface is stale. The point of the calculus is exactly the reviewer's concern: it is structurally impossible for a [C] claim to be silently rendered as [E], because the marker is read from the ledger, not retyped per document.

3 Layer 2 — no free pattern, and the reverse audit

Anti-fitting: an identity earns its keep or it is decorative

The *forward discipline* / generator-economy audit ([`verification/v305_witness_independence.py`], null model [`verification/v100_numerology_null_mc.py`]) admits an identity as load-bearing only if it is derivationally necessary, kills alternatives, is ablation-relevant, links modules, or is experimentally testable; admissible invariant classes are bounded in advance. The complementary *reverse audit* (`E8.REVERSE.AUDIT.01`) asks the honest opposite question — how much E_8 structure maps onto *nothing*? Of the eight Casimir degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$, exactly **three** feed a primary physical readout (degree 2, the metric; degree 8, the rank $\to c_3 = 1/(8\pi)$; degree 30, the Coxeter number $\to g_{\text{car}} = 5$ and the order-30 clock) and **five** carry no primary readout. A theory that publishes its 5/8 unused overhead is not cherry-picking — and those five are not diffuse slack but the forced two-family ladder $6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup \text{det-ladder}\{8, 14, 20\}$ (v431), so even the overhead is structured, not mined.

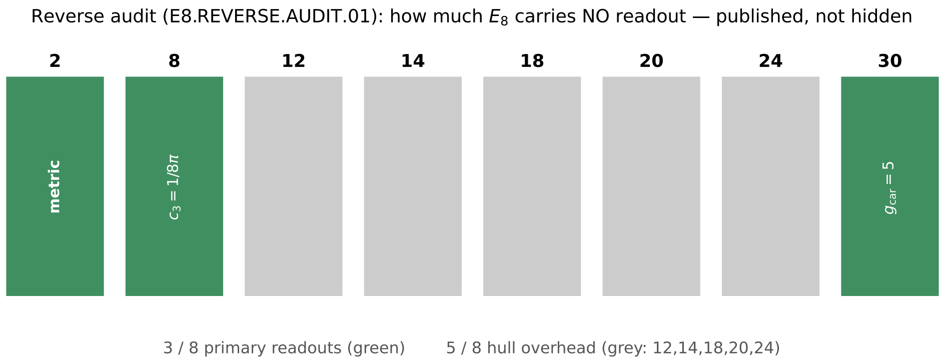

Figure 2: The reverse audit (`E8.REVERSE.AUDIT.01`): of E_8's eight Casimir degrees only three feed a primary readout; the other five are published hull overhead.

4 Layer 3 — the over-determination map (multiply vs. compress)

The strongest positive evidence is over-determination, but it must be counted honestly. The *framework* is a sharp axis ([`verification/v427_overdet_witness_map.py`], extending the generator-economy audit v305): evidence *multiplies* only across genuinely disjoint grammars; many readouts from one generator are a *compression* gain, not independent multiplication. The hard part is classifying TFPT's own witnesses correctly — and applying that test to ourselves ([`verification/v428_overdet_reclass.py`]) forces a self-correction that we state in full.

Seven arithmetic readouts — and why they *compress*

Each integer below is computed *from its own grammar* and lands on a carrier-skeleton value:

Gauss $\mathbb{Z}[i]$:	$N(3{+}2i) = 3^2{+}2^2$	$= 13\ (= \Delta_Q)$
Eisenstein $\mathbb{Z}[\omega]$:	$N(3{+}2\omega) = 3^2{-}3{\cdot}2{+}2^2$	$= 7$
Cyclotomy $\mathbb{Q}(\zeta_5)$:	$N(3{+}2\zeta_5) = 2^4\,\Phi_5(-3/2)$	$= 55$
Galois $(\mathbb{Z}/5)^\times$:	$\lvert\mathrm{Gal}\,\mathbb{Q}(\zeta_5)\rvert = \varphi(5)$	$= 4\ (= \lvert\mu_4\rvert)$
Root lattice :	$\lvert\det \mathrm{Cartan}(E_8)\rvert$	$= 1$
Pascal/exterior :	$\binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2} = 11,\ \ 2^4$	$= 16 = \dim S^+$
Coxeter :	$h(E_8) = 30 = 2{\cdot}3{\cdot}5,\ \ \varphi(30)$	$= 8 = \mathrm{rank}\, E_8$

These are seven *syntactically* distinct grammars — but syntactic distinctness is not statistical independence. By the Brieskorn classification (v236) the $(2,3,5)$ singularity $x^2{+}y^3{+}z^5$ is the *one* generator behind the order-30 clock, with Milnor number $\mu = (2{-}1)(3{-}1)(5{-}1) = 8 = \mathrm{rank}\, E_8$ and $30 = 2{\cdot}3{\cdot}5 = h(E_8)$. Each readout above is a facet of that same object: $\mathbb{Z}[i]$ is the prime-2 (μ_4/sheet) face, $\mathbb{Z}[\omega]$ the prime-3 (family) face, $\mathbb{Q}(\zeta_5)$ and the Galois group the prime-5 (carrier) face, Pascal the μ_4 exterior, the Coxeter number the whole clock, and $\det \mathrm{Cartan}(E_8)$ the E_8 resolution itself. *So the seven do not multiply — they compress*, exactly like the anchor $a = (1,1,2)$ emitting $(e_1,e_2,e_3) = (4,5,2)$, the power sums, and the E_8 data from one generator. An earlier version of this section called the seven "disjoint grammars that multiply"; [`verification/v428_overdet_reclass.py`] applies this layer's own test to that claim and walks it back.

What honestly multiplies: forced inputs + a foreign witness

The genuine over-determination is of a different shape. First, the *input* of the theory is forced by several *independent* arguments: the "8" in $c_3 = 1/(8\pi)$ equals $\mathrm{rank}\, E_8 = 8$, the carrier Coxeter number $h(D_5) = 2(5{-}1) = 8$, the totient $\varphi(30) = 8$, and the Milnor number $(2{-}1)(3{-}1)(5{-}1) = 8$ — four *different* mathematical facts that independently fix the same constant, and that genuinely multiply. Second, the constant the theory was *not* built around — $\alpha^{-1} \approx 137$, a prime foreign to the $(2,3,5)$ skeleton — is reproduced as the fixed point of the boundary Ward identity (v305). The honest claim is therefore narrower and harder to attack: not "seven independent witnesses," but *one richly over-determined input plus a foreign readout.*

The unconditional floor, and the α-form lever [C] ([`verification/v432_overdet_floor.py`])

Multiplying *only* across the genuinely disjoint pieces — the input over-determination (the "8" forced four ways), the foreign α^{-1} census ($\sim 1/94500$, v100), the φ_0-seed cluster counted *once*, and one downstream witness ($K^+ \to \pi^+ \nu\bar\nu$) — yields an *unconditional* improbability floor: $\le 10^{-6}$ across a grid of deliberately generous chance assignments, nominal $\sim 10^{-10}$. This floor is grammar-*independent* (it survives even if a skeptic rejects the declared formula grammar) and sits ~ 20 orders *above* the conditional null-model ceiling $10^{-30.7}$ (v100). That gap is exactly the evidential payoff of *deriving* the α form: the open form-derivation `ALPHA.QUILLEN.EXACT.01` (v382; its cubic level now the *computed* bulk Chern invariant $|C|{=}1$ and its normalisation the embedding index $k_Y{=}5/3$, v470; the determinant-line bridge lemma exhibited at the finite level over the $U(1)$-twist moduli, v472 — the continuum ζ-det identification still [O]) and this floor are the *same lever* — closing it moves the α factor from the $1/94500$ census toward certainty. Honest scope: this is a conservative *floor* (an upper bound on chance-plausibility), not a posterior, and it bounds only the "is it numerology?" question — not the physics bridge (the firewall, v187). *Hardened ([`verification/v436_overdet_floor2.py`]).* The verdict does not even need the chance assignments above: stripped to a *pure counting* census — of the 94500 complexity-matched $F_{U(1)}$ variants only the TFPT one hits the CODATA window (v100) — the floor is $1/94500 \approx 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar. A monotone concession ladder ($30.7 \to 25.8 \to 4.98$ dex) keeps the "not chance" verdict at the *most adversarial* rung, so it is independent of the contestable input-"8" independence; the only honest gap to the conventional 5σ line is exactly one further independent confirmation (a factor ≤ 0.054), stated openly.

5 Layer 4 — the firewall and the No-Unit theorem

Two safeguards keep the boundary between the closed core and external physics sharp.

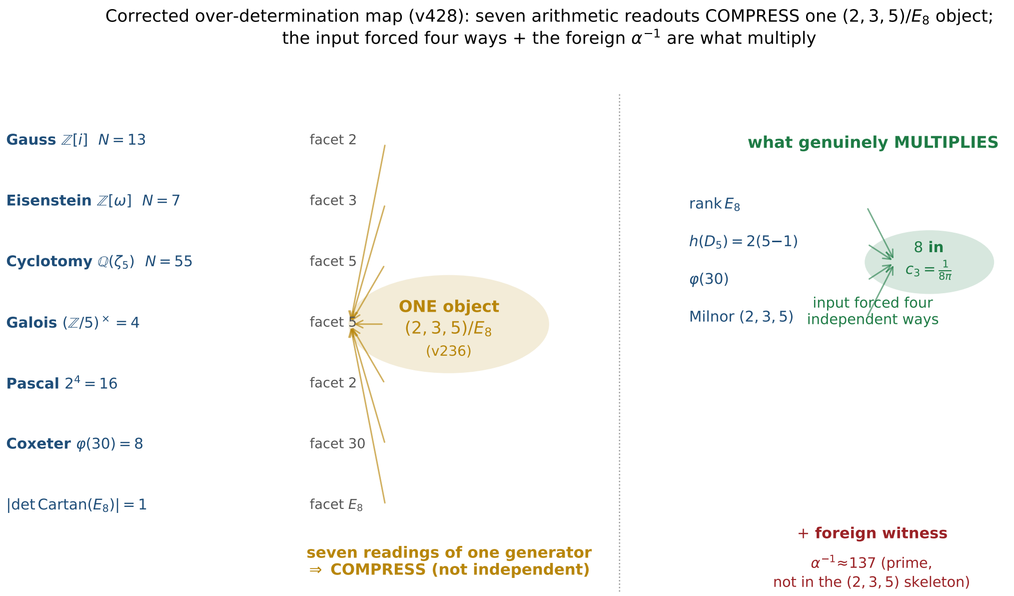

Figure 3: The corrected over-determination map ([verification/v428_overdet_reclass.py]): the seven arithmetic readouts are facets of *one* $(2,3,5)/E_8$ object (a *compression* gain, like the anchor), so they do not multiply; what genuinely multiplies is the input forced four independent ways (the "8") plus the foreign witness $\alpha^{-1}\approx 137$. The disjoint pieces give an *unconditional* floor $\sim 10^{-10}$ ([verification/v432_overdet_floor.py]), ~ 20 orders above the conditional $10^{-30.7}$; hardened to an assumption-minimal *counting* floor $1/94\,500 \approx 4.40\sigma$ ([verification/v436_overdet_floor2.py]) that drops every subjective chance assignment and a monotone concession ladder ($30.7 \to 25.8 \to 4.98$ dex) that keeps the "not chance" verdict at the most adversarial rung.

- **The firewall.** The four frontier transfers — F_{pole} (Koide), $F_{\text{Boltzmann}}$ (η_B), F_{relic} (axion), F_{QCD} (m_p/m_e) — are *typed interfaces*, a discrete compiler kernel composed with an external continuous solver. A guard ([verification/v187_ftransfer_laws.py]) reads the ledger and *enforces* that none is ever typed as a primitive [**E**] compiler prediction; the functor contract is [verification/v213_ftransfer_functor.py] and a prose sentinel ([verification/v188_frontier_wording_guard.py]) guards the wording. Exact algebraic sub-parts may be [**E**] (the Koide 53/54 factor, $b_3=-7$); the physical observable never is. The recent single-flow reduction ([verification/v425_dyn_transfer_universal.py]) makes the *dynamics* of all four the one native seam recovery semigroup, leaving only named anchors external — a sharpening that respects, not relaxes, the firewall.
- **The No-Unit theorem** ([verification/v153_no_unit_theorem.py]). Because c_3 and g_{car} are mass-dimension 0, no dimensionless compiler can output an absolute scale; v_{geo} is therefore *forbidden by theorem*, not a missing prediction. The residual-certification audit ([verification/v384_residual_certification.py]) makes the shape explicit: every open item is an external math proof, theorem-forbidden (the unit), or external physics — *zero* open internal mechanisms.

6 Layer 5 — frozen predictions, the null model, and the α uniqueness scan

Predictions are frozen before confrontation, and a match is scored against *chance*, not admired in isolation.

The look-elsewhere null model, made explicit ([`verification/v100_numerology_null_mc.py`])

The frozen registry ([`verification/v84_frozen_registry.py`], `REG.FREEZE.01`) fixes every dimensionless prediction at 25 digits, re-derived from the two axioms each run (a formula↔value lock). The null model then runs an *exact census* of the full declared formula grammar over the 13 scored observables with conservative data windows: the joint formula-fishing probability is $\prod_i p_i = 10^{-25.8}$, and 2×200,000 Monte-Carlo pseudo-theories reach *at most* 5/13 hits against TFPT's 13/13 (Fig. 4). Power controls confirm the test bites: $\varphi_0^{\text{ret}} \pm 10\%$ collapses TFPT's own scorecard to 6/13, and a data shuffle averages 1.2/13.

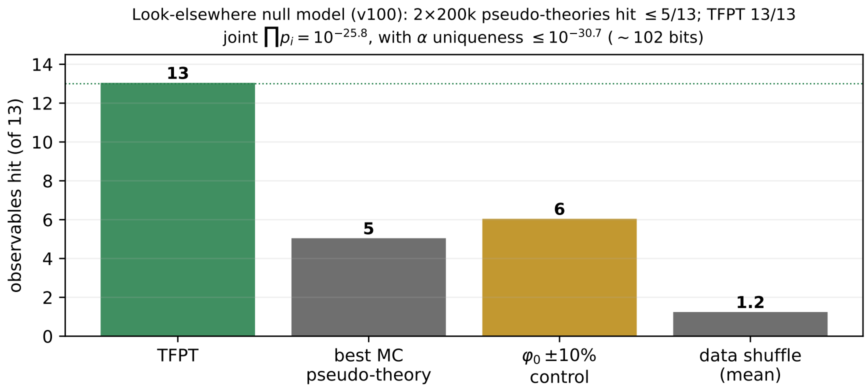

Figure 4: The look-elsewhere null model ([`verification/v100_numerology_null_mc.py`]): Monte-Carlo pseudo-theories hit at most 5/13 where TFPT hits 13/13; the power controls ($\varphi_0^{\text{ret}} \pm 10\%$, data shuffle) confirm the test is non-vacuous.

Is the α path unique? The sharpest single result is the fine-structure fixed point. All 94,500 variants of the boundary $U(1)$ Ward formula $F_{U(1)}$ in the declared grammar are root-solved; *exactly one* — the budget $M{=}41$ at $N{=}8$ ($c_3 = 1/8\pi$) — lands in the $4{\times}10^{-8}$ CODATA window (Fig. 5). Folding this into the census gives a total $\leq 10^{-30.7}$ (∼102 bits), *conditional on the declared grammar* — a null-model rejection, never a claim of certainty.

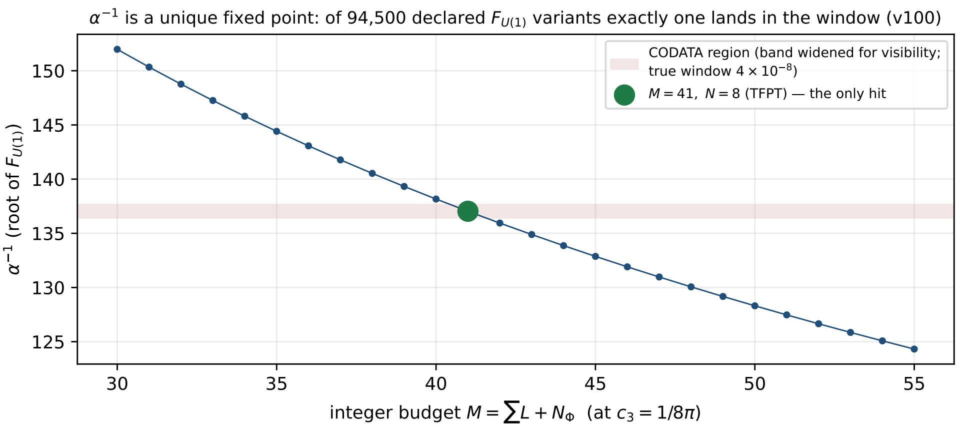

Figure 5: α^{-1} as a unique fixed point: scanning the integer budget M at $c_3 = 1/8\pi$, only $M{=}41$ lands in the CODATA window; the full census over all 94,500 $F_{U(1)}$ variants finds exactly one hit.

No moving goalposts; the worst case is shown. A live data scorecard records the JUNO/NuFIT/ACT/BK18 compatibility ([`verification/v375_observatory_registry.py`]). A data watchdog types each row as PASS, WATCH, TENSION or KILL ([`verification/v307_data_watchdog.py`]), and a kill-test board keeps the falsifiers tied to the frozen rows ([`verification/v321_killtest_board.py`]).

The discipline is applied to its own weakest point: the reactor angle $\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}}\, e^{-5/6}$ is *pre-registered* at $+2.0\sigma$ tension (`FLAV.TH13.PRESSURE.01`) with its own kill criterion — the opposite of hiding the worst case.

7 Layer 6 — two independent reproduction paths

A single implementation can share a single bug; TFPT therefore re-derives the exact core twice more, independently.

- **An independent Wolfram engine** (`GATE.WOLFRAM.01`/`GATE.WOLFRAM.02`): 116/116 base checks plus a 588/588 extension mirror every exact algebraic / identity / lattice result in a different computer-algebra system, ending `ALL WOLFRAM CHECKS PASSED`. (Numerical ODE/PDE results are Python-only by design and flagged as such.)
- **A Lean 4 kernel proof**: the load-bearing algebra is machine-formalised with no `sorry` — the carrier hypercharge as the exact anomaly-free root of $6Y^2 - Y - 1 = 0$, the Pascal carrier ladder $2^4 = \binom{5}{0}+\binom{5}{1}+\binom{5}{2}$ (`FORM.LADDER.01`), the integer rigidity of the carrier, the seam-equivalence chain (`FORM.SEAMEQUIV.01`) and its collar/MMST face (`FORM.SEAM.MMST.01`) — so the kernel, not the prose, certifies the core.

8 Layer 7 — the red team

The adversarial audit (companion `tfpt_5_redteam`) does not defend; it *attacks*. It declares six targets — A (the seam $=(E_8)_1$ identification), B (the carrier truncation $g_{\mathrm{car}}{=}5$), C (the $k{=}c_3/2$ area law), D (the ratios $\to v_{\mathrm{geo}}$ reduction), E (whether one scale carries the theory), F (the perturbative 4D-QFT round) — states what survives each, and reduces every survivor to a named residual. The model example is its own honesty about holomorphy: $c = 8$ alone does *not* select $(E_8)_1$ because $(D_8)_1 = SO(16)_1$ shares $c = 8$; the load-bearing extra is holomorphy ($|\det K| = 1$). Publishing the sharpest attack on oneself is the strongest anti-numerology signal a programme can give.

9 What would still falsify TFPT

The safeguards are only credible with explicit kill tests (companion "How to kill TFPT"). Among the frozen ones: a fourth chiral generation (breaks $N_{\mathrm{fam}} = 3$); a genuinely free fundamental constant that is not an E_8-closure fixed point; a seam denominator $\neq 8$; α^{-1} away from the fixed-point root; a stable $\sin^2\theta_{13}$ far from 0.0231 at $> 3\sigma$; cosmic birefringence converging away from $\beta \approx 0.2424°$; a tensor-to-scalar $r \gtrsim 0.01$ (which kills the R^2 branch). The honest residual above all kill tests is exactly three named interfaces — the seam continuum theorem (`SEAM.EQUIV.01`, now reduced to a cited commuting-projector result, [`verification/v424_seam_lto_rp_reduction.py`]/[`verification/v426_seam_lto_rp_kms.py`], its extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input at invariant level, [`verification/v469_seam_crossedproduct_route.py`]), the metrology unit v_{geo} (theorem-forbidden), and the typed F_{transfer} interfaces (firewalled).

10 Summary — the layered defense

Layer	What it defends against	Artefact
Status calculus	a [C] claim read as [E]	ledger + `audit_sync.py`
No free pattern + reverse audit	cherry-picking, decorative identities	v305/v100/v431/`E8.REVERSE.AUDIT.01`
Over-determination map	rhetorical "everything multiplies" (incl. *our own* miscount)	v427/v428/v432/v436
Firewall + No-Unit	external physics sold as compiler output; a faked scale	v187/v213/v153/v384
Frozen registry + null model	moving goalposts; chance hits	v84/v100/v375
Two independent paths	a shared implementation bug	Wolfram + Lean
Red team	self-serving framing	`tfpt_5_redteam`

The claim defended here is deliberately narrow: *coincidence is an expensive explanation of the discrete core, and exact compiler closure is never silently upgraded to physical closure.* That is the whole job of this companion.